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CS23633 – Cloud Computing Laboratory

Experiment 5: Cloud Migration – Backup Setup experiment using Google Drive as the cloud storage service.

Aim

To create a cloud backup of local files using Google Drive, enable automatic synchronization through Google Drive for Desktop, and verify that changes made to a local file are reflected in the cloud.

PART A — Creating the Sample Files

Step 1: Create a folder

1. Open the **Desktop** on the computer.
2. Right-click on an empty area.
3. Select **New** → **Folder**.
4. Name the folder as:

OfficeData

Step 2: Create text files

Open the **OfficeData** folder and create the following three text files:

- Student.txt
- Attendance.txt
- Marks.txt

Enter the following information in each file.

Student.txt

Name : Student

Department : AI & DS

Attendance.txt

Attendance : 95%

Marks.txt

Cloud Computing : 98

The folder structure should be:

OfficeData

├── Student.txt

├── Attendance.txt

└── Marks.txt

PART B — Setting Up Cloud Storage

Step 3: Open Google Drive

1. Open a web browser.
2. Visit **Google Drive**.
3. Sign in using your Google account.

Step 4: Create a cloud backup folder

1. Click the + **New** button.
2. Select **New Folder**.
3. Enter the folder name:

CloudBackup

4. Click **Create**.

PART C — Uploading Files to Google Drive

Step 5: Upload the sample files

1. Open the newly created **CloudBackup** folder.
2. Click + **New**.
3. Select **File upload**.
4. Choose the following files:
 - Student.txt
 - Attendance.txt
 - Marks.txt
5. Click **Open**.

After the upload is completed, the Google Drive folder should contain:

CloudBackup

|— Student.txt
|— Attendance.txt
|— Marks.txt

This step transfers the locally created files to the cloud storage provided by Google Drive.

PART D — Installing Google Drive for Desktop

Step 6: Install Google Drive for Desktop

1. Download **Google Drive for Desktop** from the official Google Drive website.
2. Install the application on the computer.
3. Launch Google Drive for Desktop.
4. Sign in with the **same Google account** used for Google Drive.

PART E — Configuring Automatic Synchronization

Step 7: Add the OfficeData folder

1. Open **Google Drive for Desktop**.
2. Open **Settings / Preferences**.
3. Select **Add folder**.

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4. Navigate to:

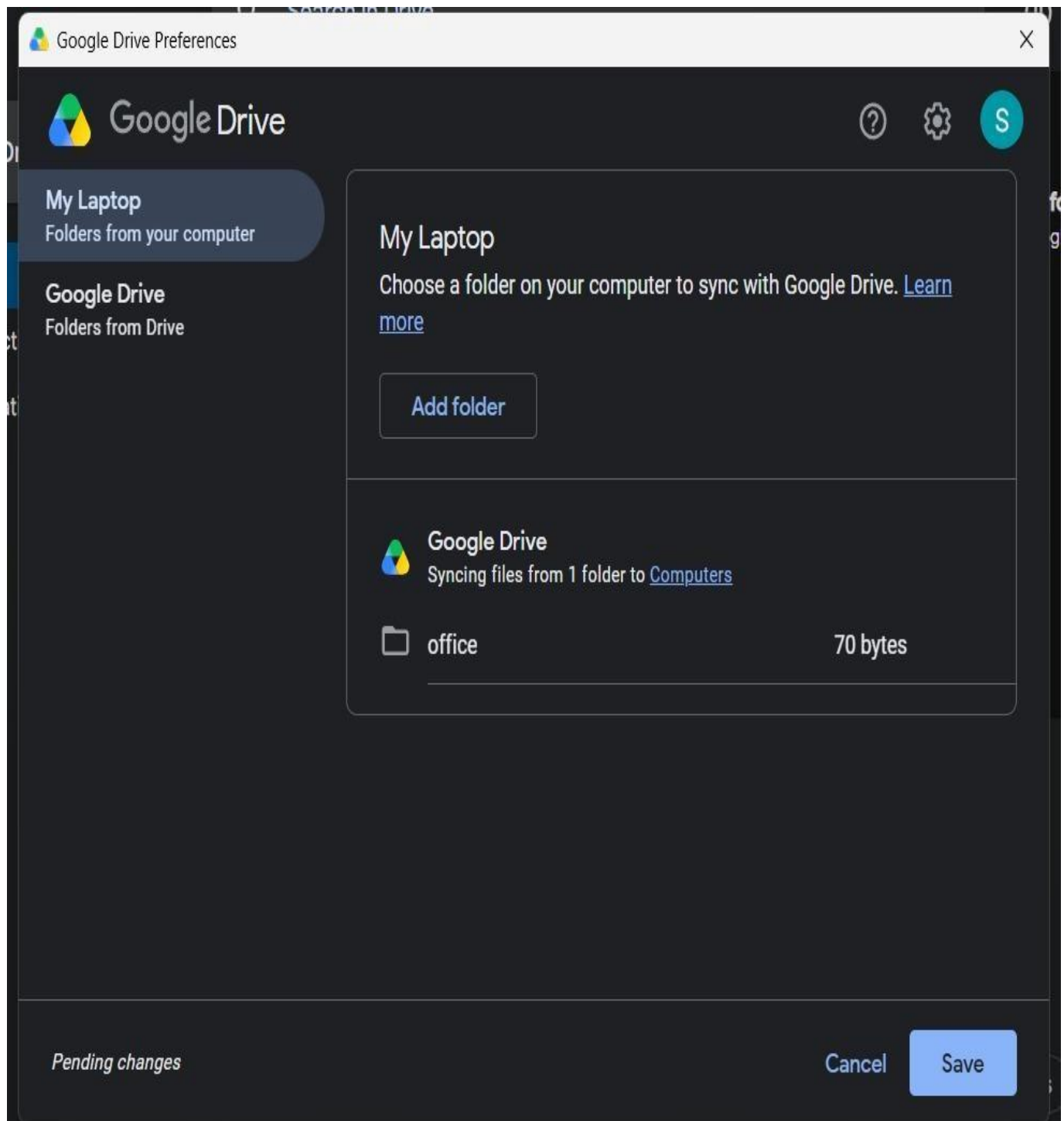
Desktop → OfficeData

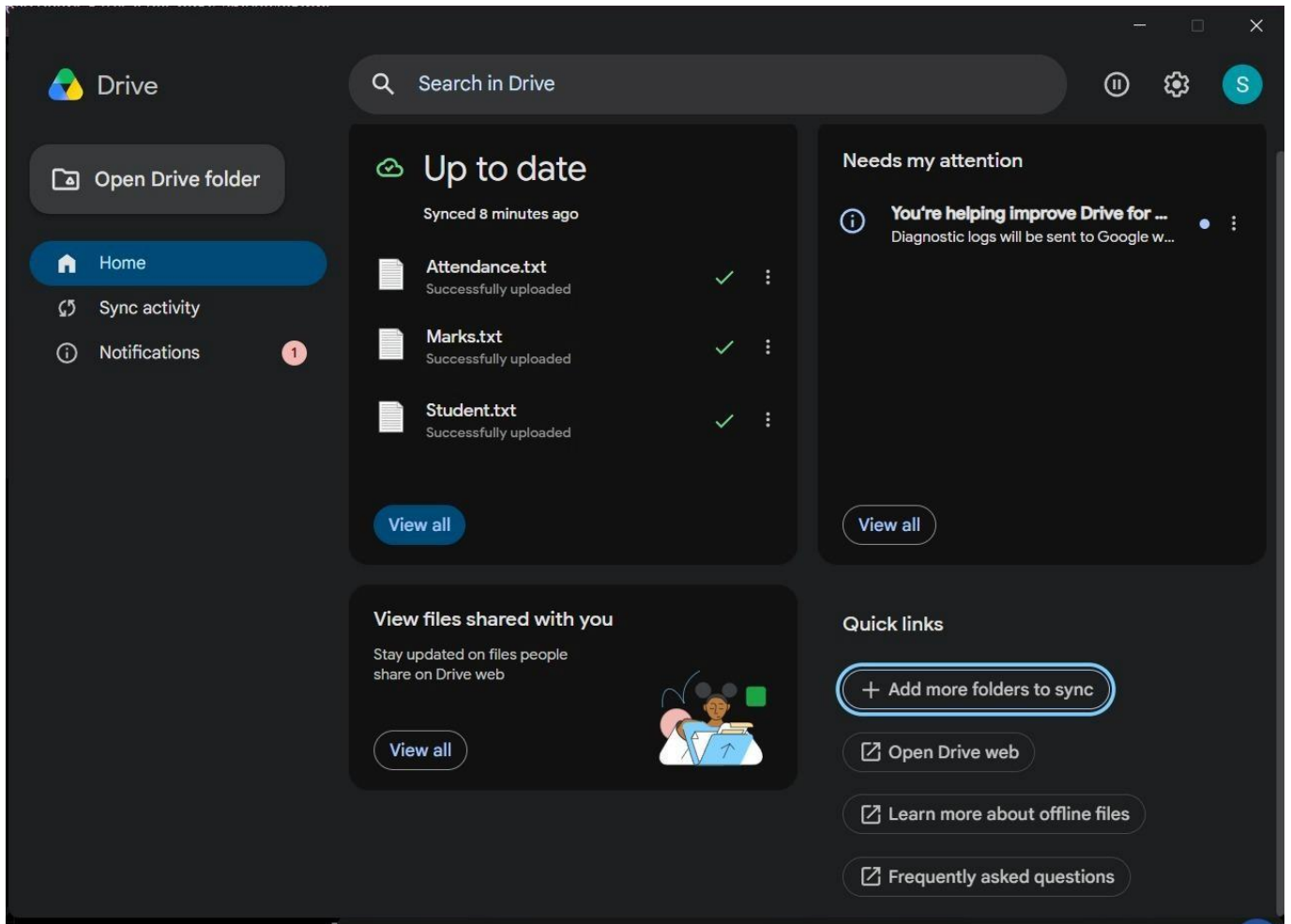
5. Select the **OfficeData** folder.

6. Enable **Sync with Google Drive**.

7. Click **Done**.

The selected folder will now be synchronized with Google Drive. Therefore, changes made to files inside OfficeData can be automatically reflected in the cloud.





PART F — Testing the Automatic Backup

Step 8: Verify the synchronized files

1. Open Google Drive in the browser.
2. Locate the synchronized **OfficeData** folder.
3. Check whether the following files are available:

OfficeData

- |— Marks.txt
- |— Attendance.txt
- |— Student.txt

Step 9: Modify Student.txt

1. Open the **OfficeData** folder on the computer.
2. Open Student.txt.
3. Add the following line at the end of the file:

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Cloud Backup Test - Successfully Synchronized

4. Save the file.

Step 10: Allow synchronization

Wait for a short period, approximately **10–30 seconds**, so that Google Drive for Desktop can synchronize the modified file.

Check Google Drive for Desktop and make sure the synchronization status indicates that the files are **up to date** or successfully synchronized.

Step 11: Verify the updated file in Google Drive

1. Open Google Drive in the browser.
2. Refresh the page.
3. Open the synchronized Student.txt file.
4. Verify that the newly added line is present.

The file should contain:

Name : Student

Department : AI & DS

Cloud Backup Test - Successfully Synchronized

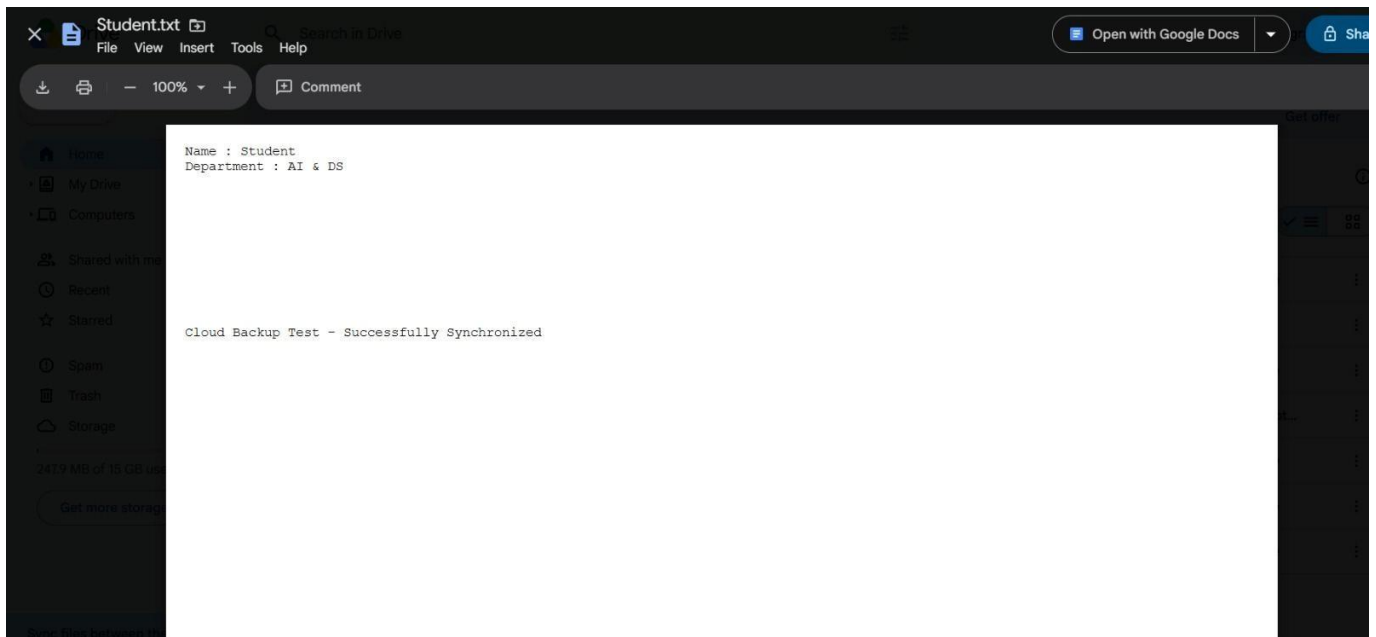
The appearance of the newly added line in Google Drive confirms that the modification made to the local file has been successfully synchronized with the cloud.

The screenshot shows the Google Drive web interface. The left sidebar contains navigation options: Home, My Drive, Computers, Shared with me, Recent, Starred, Spam, Trash, and Storage. The main area displays 'Welcome to Drive' with 'Suggested folders' (cloudbackup, Colab Notebooks) and 'Suggested files'. A table lists the files:

Name	Reason suggested	Owner	Location
Student.txt	You uploaded • 6:46 PM	me	cloudbac...
Marks.txt	You uploaded • 6:46 PM	me	cloudbac...
Attendance.txt	You created • 6:46 PM	me	cloudbac...
AWS.pdf	You opened • Jul 30	me	My Drive
sample project 1.mp4	You opened • Sep 11, 2025	uiux.internez...	Day 05
AWS.pdf	You opened • Jul 28	me	My Drive
Untitled1.ipynb	You edited • Aug 18	me	Colab Not...

A blue banner at the bottom left states: 'Sync files between the cloud and your computer. Learn more' with a 'Download' button.

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FLOW OF THE EXPERIMENT

Create OfficeData folder → Create text files → Create CloudBackup folder → Upload files → Install Google Drive for Desktop → Add OfficeData folder → Enable synchronization → Modify Student.txt → Wait for synchronization → Verify the updated file in Google Drive

Result / Conclusion

Thus, the local files were successfully uploaded to Google Drive and synchronized using Google Drive for Desktop. The modification made to Student.txt was also reflected in Google Drive, demonstrating successful **cloud backup and automatic file synchronization**.